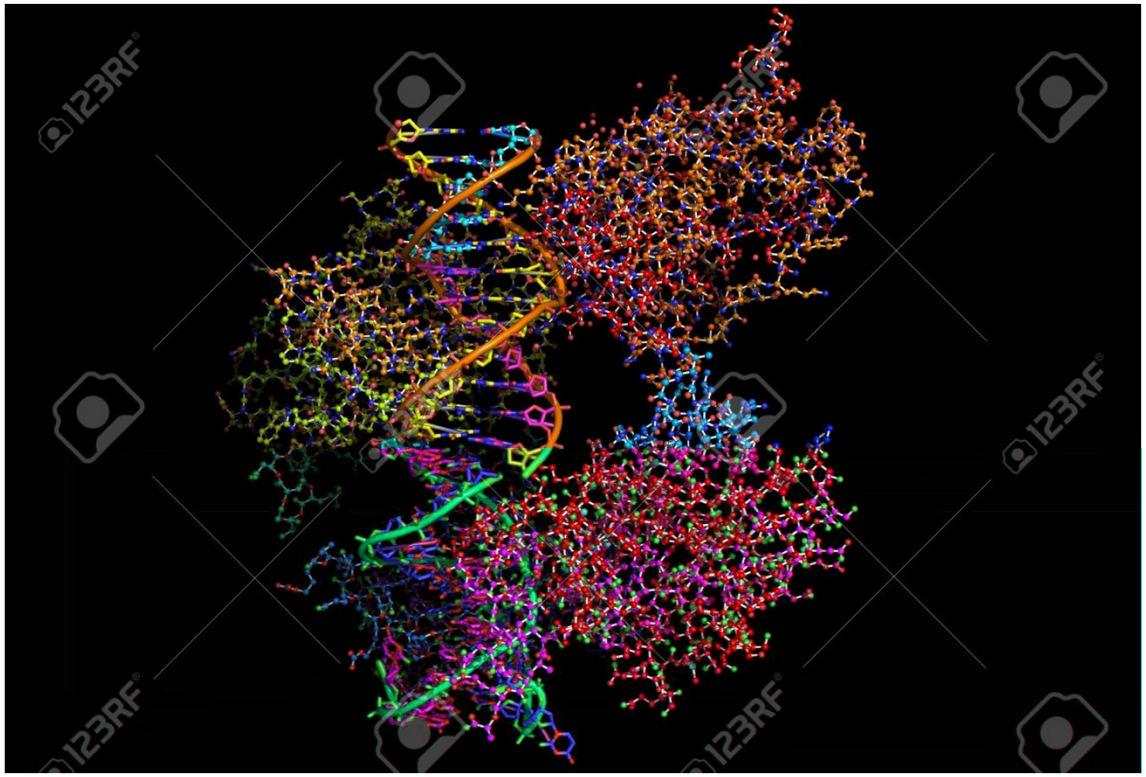
The Three-Dimensional Structure of Proteins



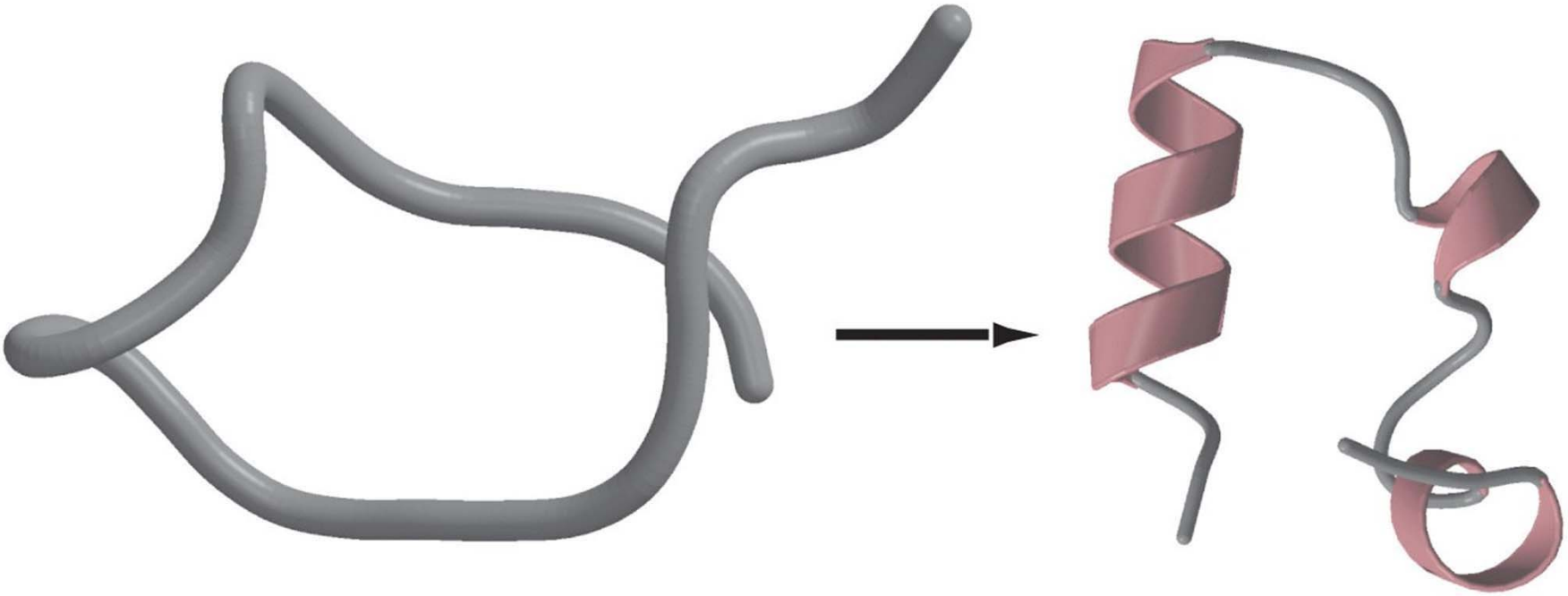
A ball and stick model of molecules of protein p53 binding to a strand of DNA. The majority of human cancers involve mutations which make this protein inactive

The Three-Dimensional Structure of Proteins

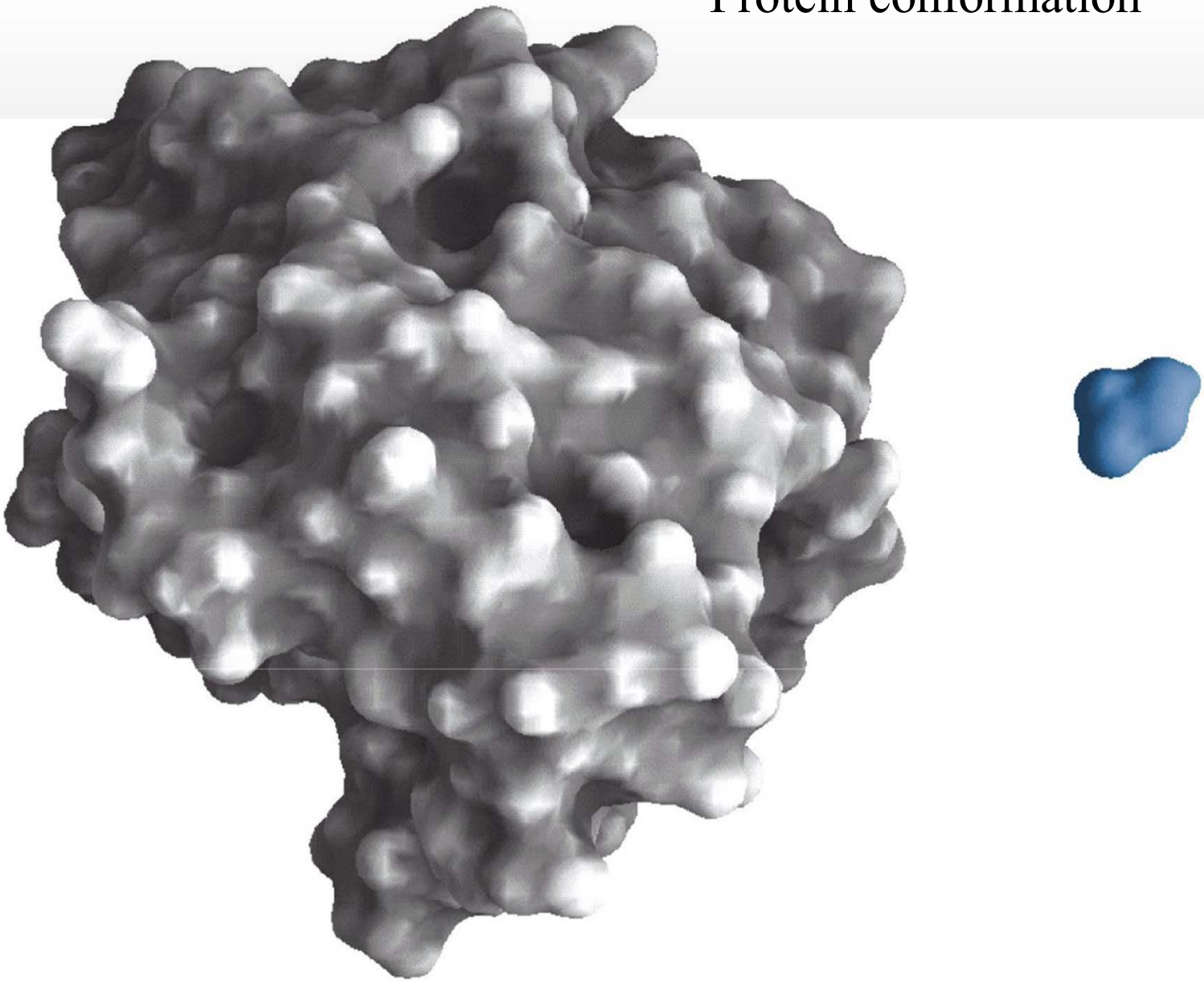
Learning goals:

- Structure and properties of the peptide bond
- Structural hierarchy in proteins
- How proteins fold and how they maintain their folding
- Structure and function of fibrous proteins
- Structure analysis of globular proteins
- Protein denaturation

Proteins have **specific** tridimensional structures



Protein conformation

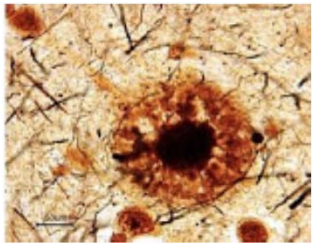


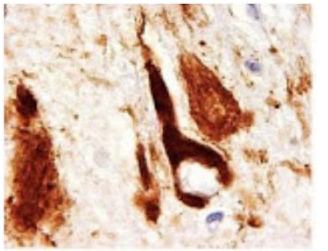
A common features in neurodegenerative diseases

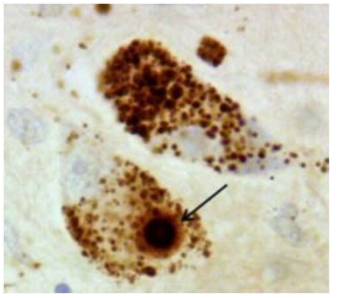
Universidad de Navarra

Abnormal protein accumulation is a common feature in neurodegenerative diseases

Protein	Disease
β-amyloid	ALZHEIMER
Tau	ALZHEIMER PICK'S DISEASE FRONTOTEMPORAL DEMENTIA PROGRESSIVE SUPRANUCLEAR PALSY
α-synuclein	PARKINSON'S DISEASE







Structure of Proteins

- Unlike most organic polymers, protein molecules adopt a specific three-dimensional conformation.
- This structure is able to fulfill a specific biological function.
- This structure is called the native fold.
- The native fold has a large number of favorable interactions within the protein.

Four Levels of Protein Structure

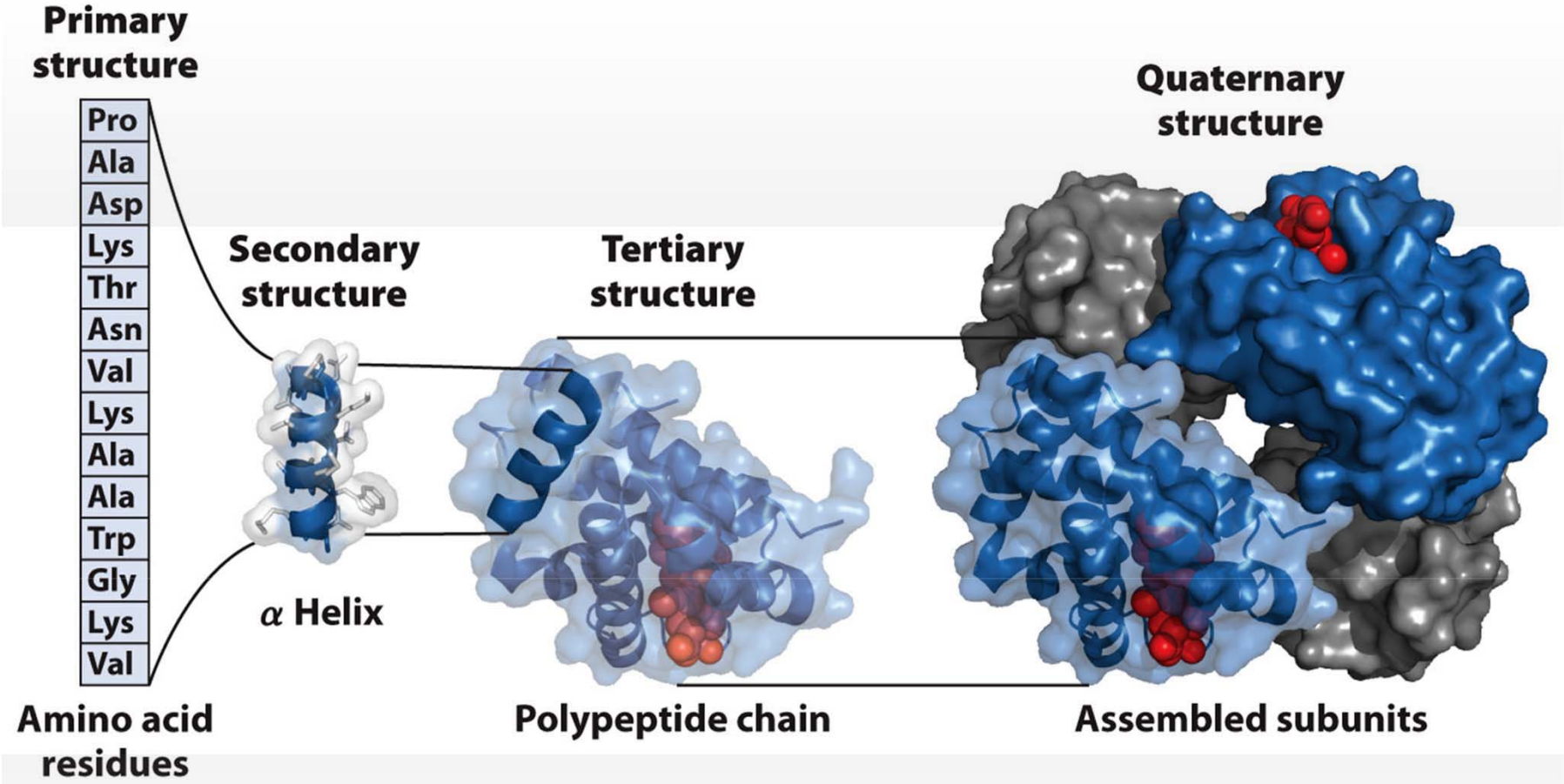


Figure 3-23
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Favorable Interactions in Proteins

- **Hydrophobic effect**
  - The release of water molecules from the structured solvation layer around the molecule as protein folds increases the net entropy.
- **Hydrogen bonds**
  - Interaction of N−H and C=O of the peptide bond leads to local regular structures such as α helices and β sheets.
- **Electrostatic interactions**
  - long-range strong interactions between permanently charged groups
  - Salt bridges, especially those buried in the hydrophobic environment, strongly stabilize the protein.
- **Van der Waals forces**
  - Medium-range weak attraction between all atoms contributes significantly to the stability in the interior of the protein.

Primary Structure: The Peptide Bond

- The structure of the protein is partially dictated by the properties of the peptide bond.
- The peptide bond is a resonance hybrid of two canonical structures.
- The resonance causes the peptide bonds to be quite **rigid** and nearly **planar**

Amino Acids Polymerize to Form Peptides

- Peptides are small condensation products of amino acids.
- They are "small" compared with proteins (M_w < 10 kDa).

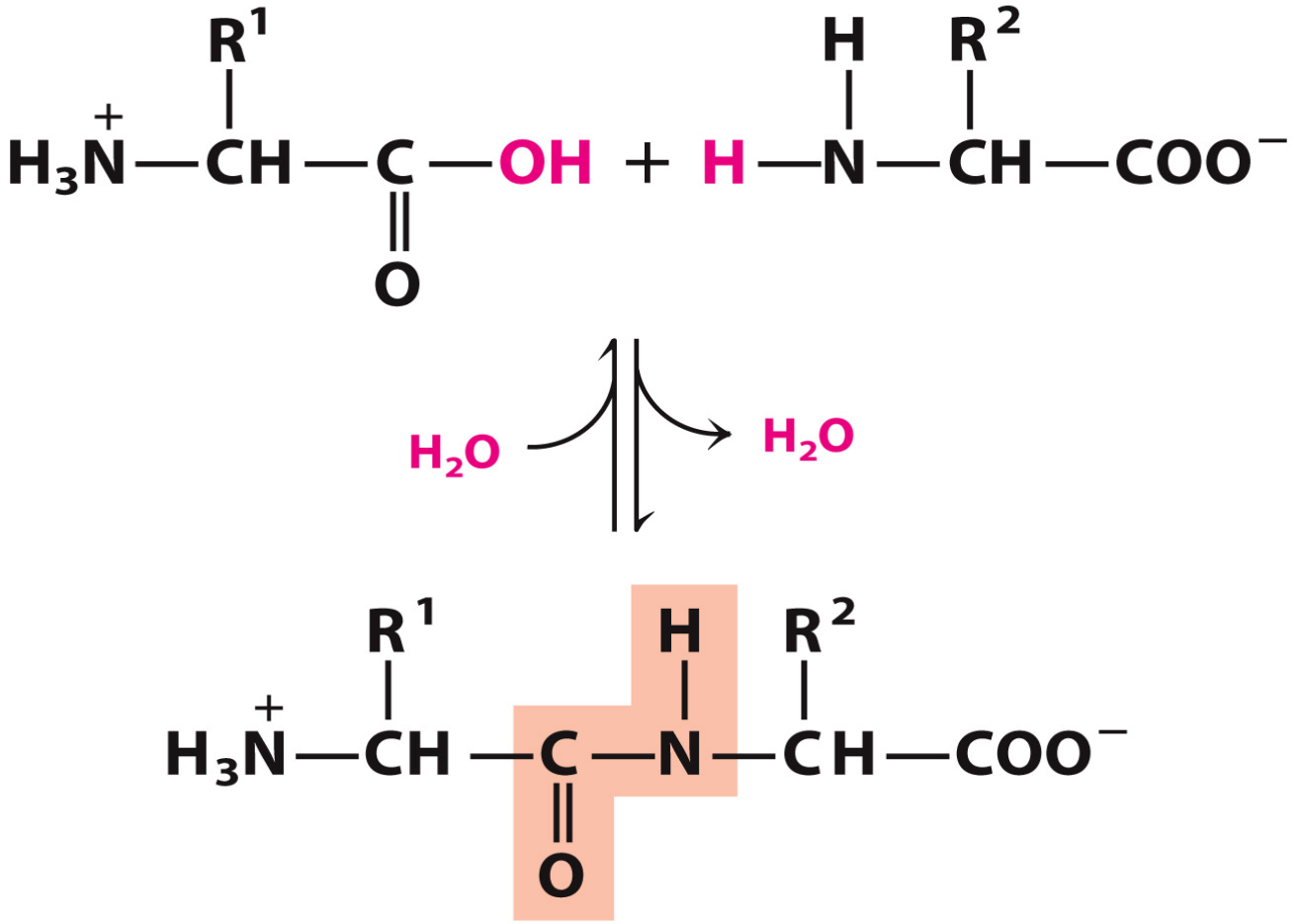


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Peptide Ends Are Not the Same

Numbering (and naming) starts from the amino terminus (N-terminal).

AA_1 AA_2 AA_3 AA_4 AA_5

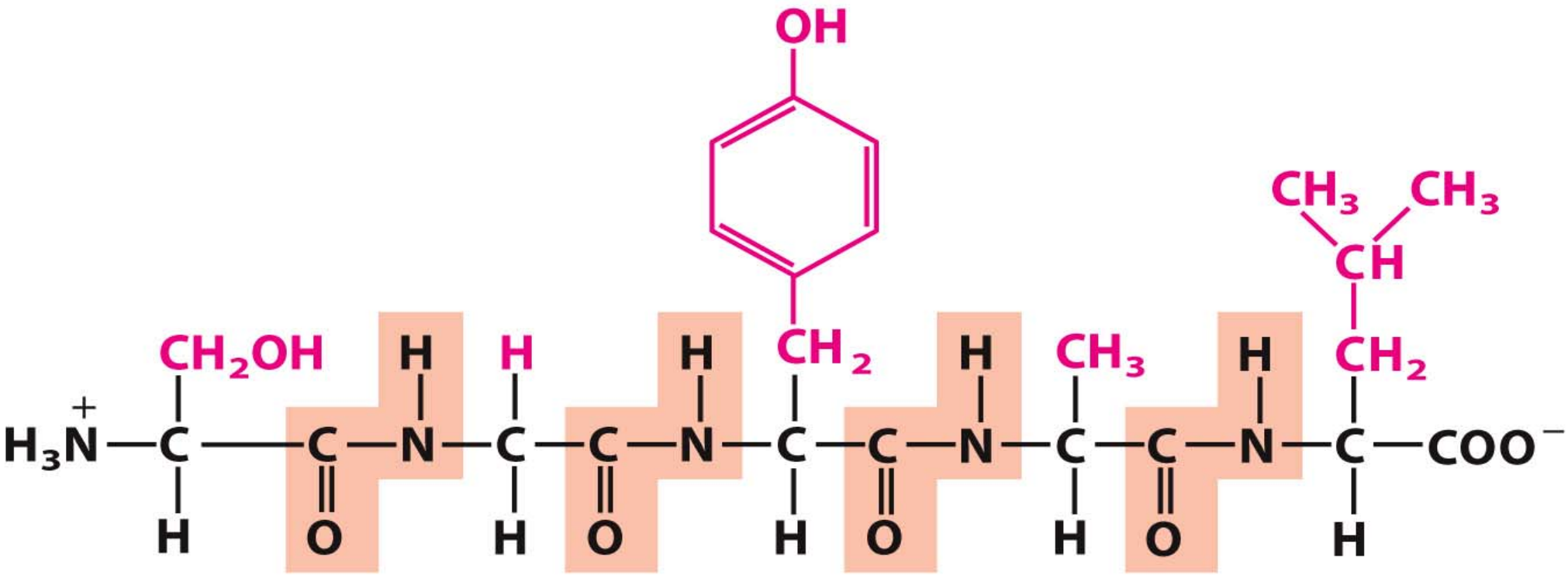


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Resonance in the Peptide Bond

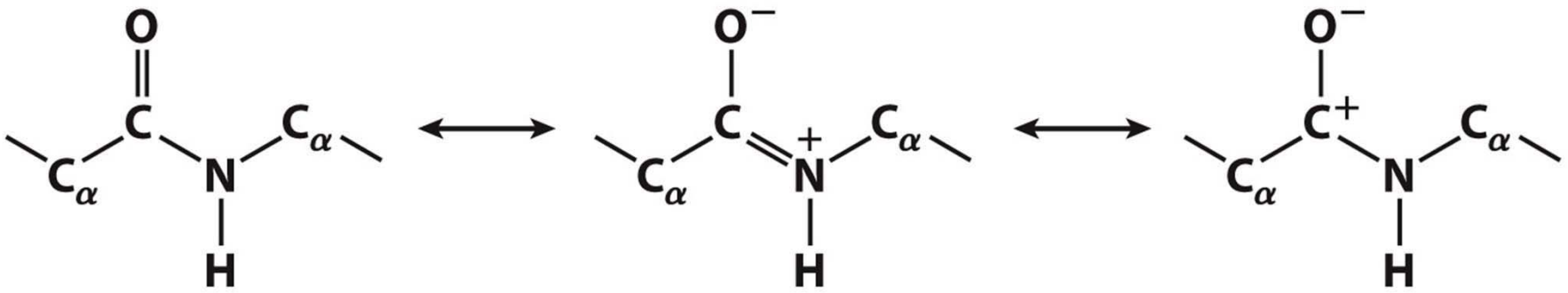


Figure 4-2a
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The Rigid Peptide Plane and the Partially Free Rotations

- Rotation around the peptide bond is not permitted due to resonance structure.
- Rotation around bonds connected to the α carbon is permitted.
  - ϕ (phi): angle around the α carbon—amide nitrogen bond
  - ψ (psi): angle around the α carbon—carbonyl carbon bond
- In a fully extended polypeptide, both ψ and ϕ are 180°.

The organization around the peptide bond, paired with the identity of the R groups, determines the secondary structure of the protein.

The Polypeptide Is Made Up of a Series of Planes Linked at α Carbons

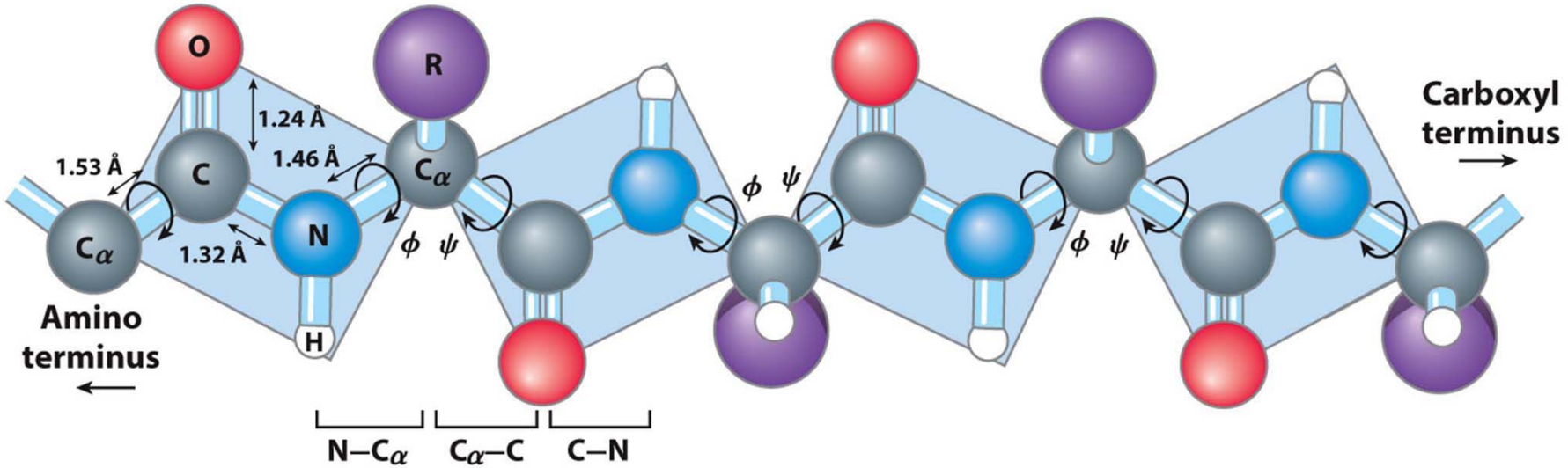
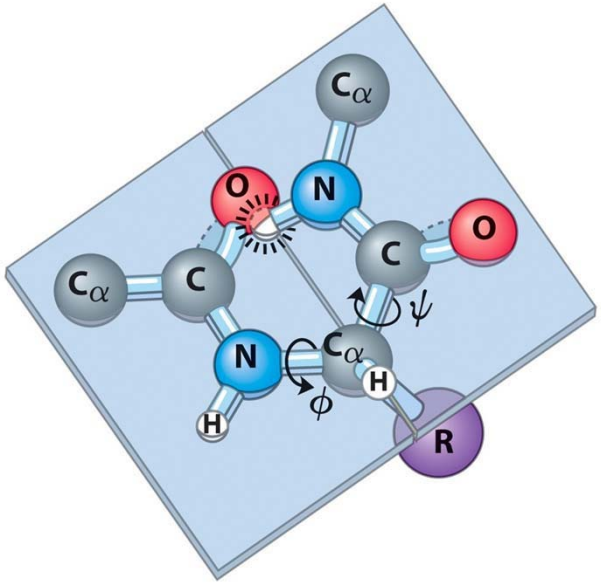


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Distribution of ϕ and ψ Dihedral Angles

- Some ϕ and ψ combinations are very unfavorable because of steric crowding of backbone atoms with other atoms in the backbone or side chains.



- Some ϕ and ψ combinations are more favorable because of chance to form favorable H-bonding interactions along the backbone.

Four Levels of Protein Structure

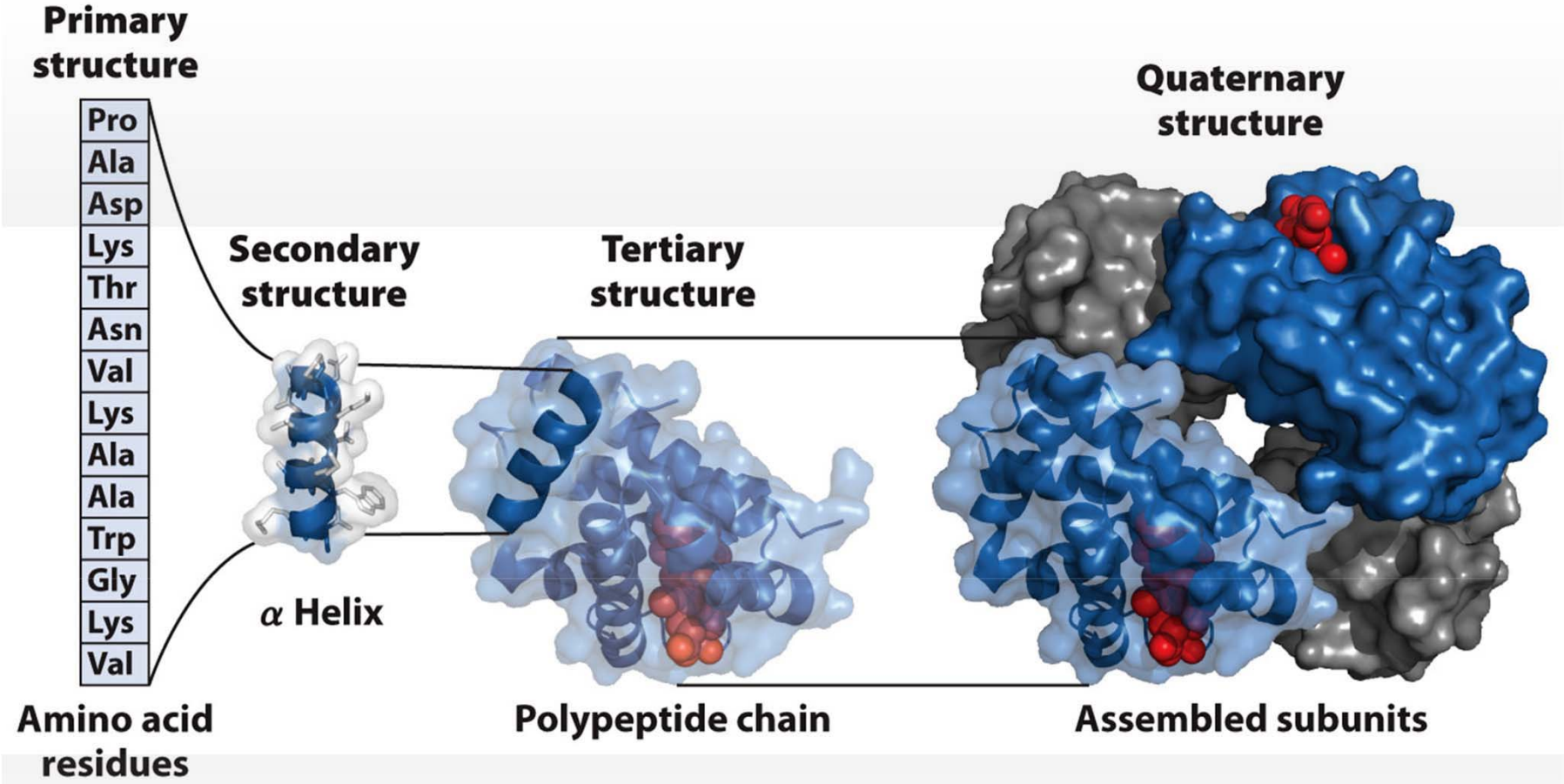


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Secondary Structures

- Secondary structure refers to a local spatial arrangement of the polypeptide backbone.
- Two regular arrangements are common:
  - the **α helix**
    - stabilized by hydrogen bonds between nearby residues
  - the **β sheet**
    - stabilized by hydrogen bonds between adjacent segments that may not be nearby
- Irregular arrangement of the polypeptide chain is called the random coil.

The α Helix

- Helical backbone is held together by hydrogen bonds between the backbone amides of an *n* and *n* + 4 amino acids.
- It is a right-handed helix with 3.6 residues (5.4 Å) per turn.
- Peptide bonds are aligned roughly parallel with the helical axis.
- Side chains point out and are roughly perpendicular with the helical axis.

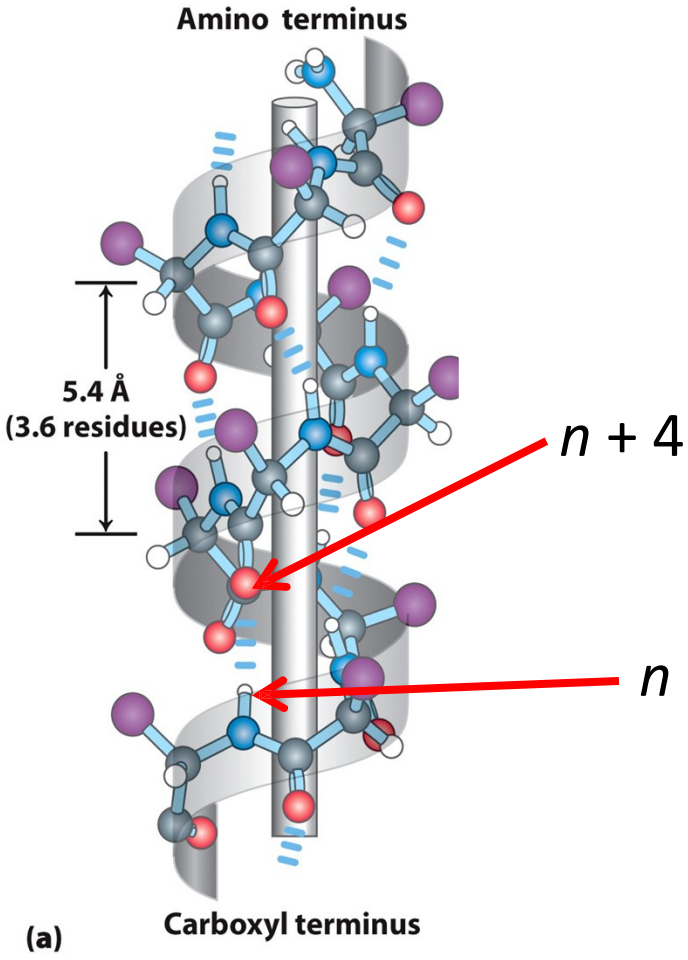


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Sequence Affects Helix Stability

- Not all polypeptide sequences adopt α-helical structures.
- Small hydrophobic residues such as Ala and Leu are strong helix formers.
- Pro acts as a helix breaker because the rotation around the N-C_a (φ-angle) bond is impossible.
- Gly acts as a helix breaker because the tiny R group supports other conformations.
- Attractive or repulsive interactions between side chains 3 to 4 amino acids apart will affect formation.

β Sheets

- The planarity of the peptide bond and tetrahedral geometry of the α carbon create a pleated sheet-like structure.
- Sheet-like arrangement of the backbone is held together by hydrogen bonds between the backbone amides in different strands.
- Side chains protrude from the sheet, alternating in an up-and-down direction.

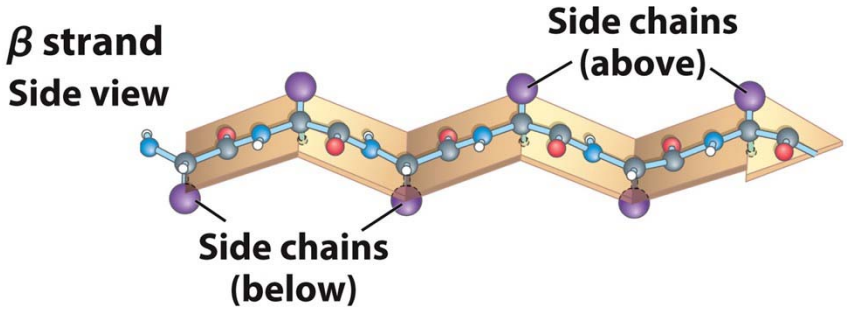


Figure 4-6a
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Parallel and Antiparallel β Sheets

- Multi β-strand interactions are called sheets.
- Sheets are held together by the hydrogen bonding of amide and carbonyl groups of the peptide bond from opposite strands.
- Two major orientations of β sheets are determined by the directionality of the strands within:
  - Parallel sheets have strands that are oriented in the same direction.
  - Antiparallel sheets have strands that are oriented in opposite directions.

In parallel β sheets, the H-bonded strands run in the same direction.

- Hydrogen bonds between strands are bent (weaker).

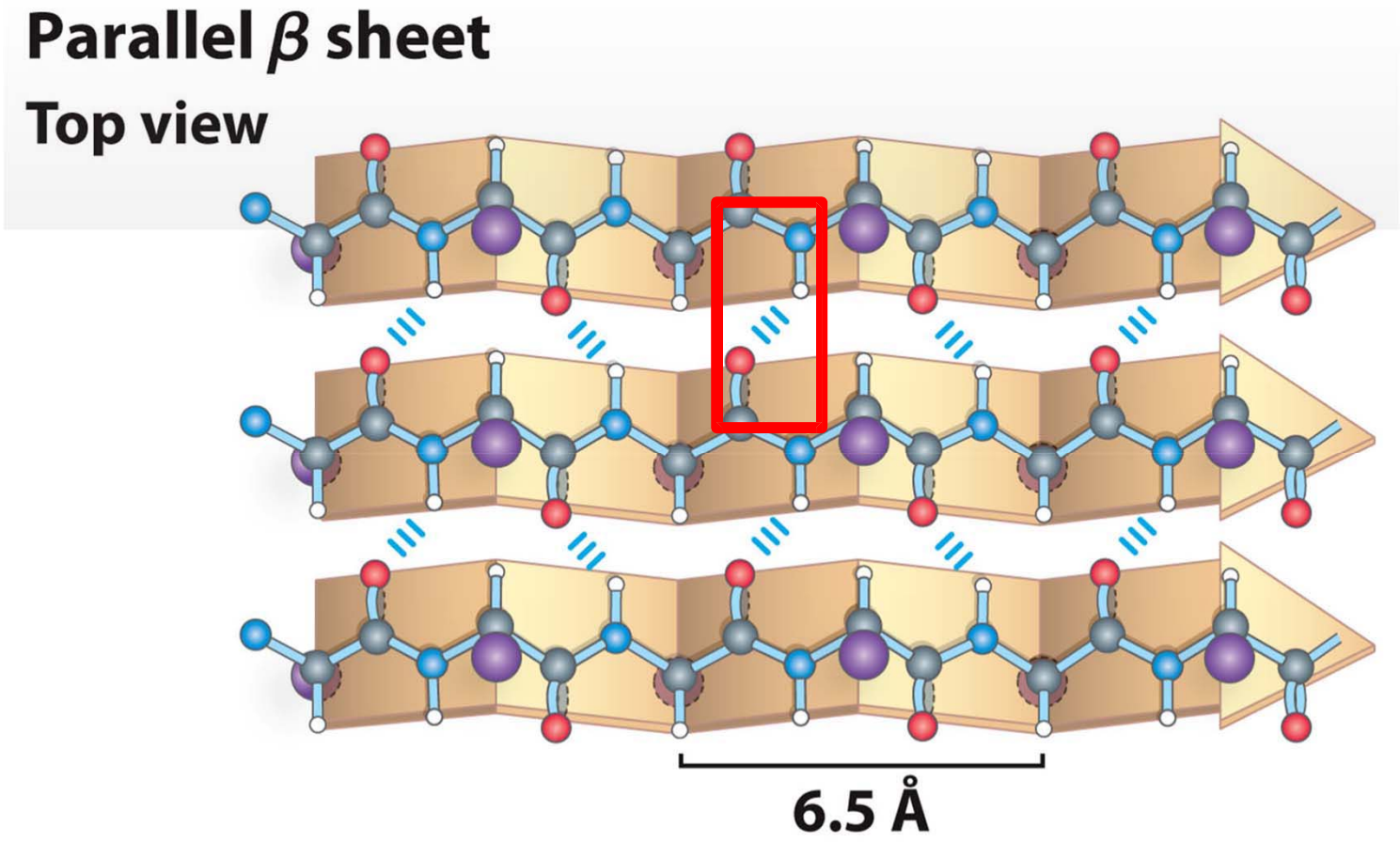


Figure 4-6c
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In antiparallel β sheets, the H-bonded strands run in opposite directions.

- Hydrogen bonds between strands are linear (stronger).

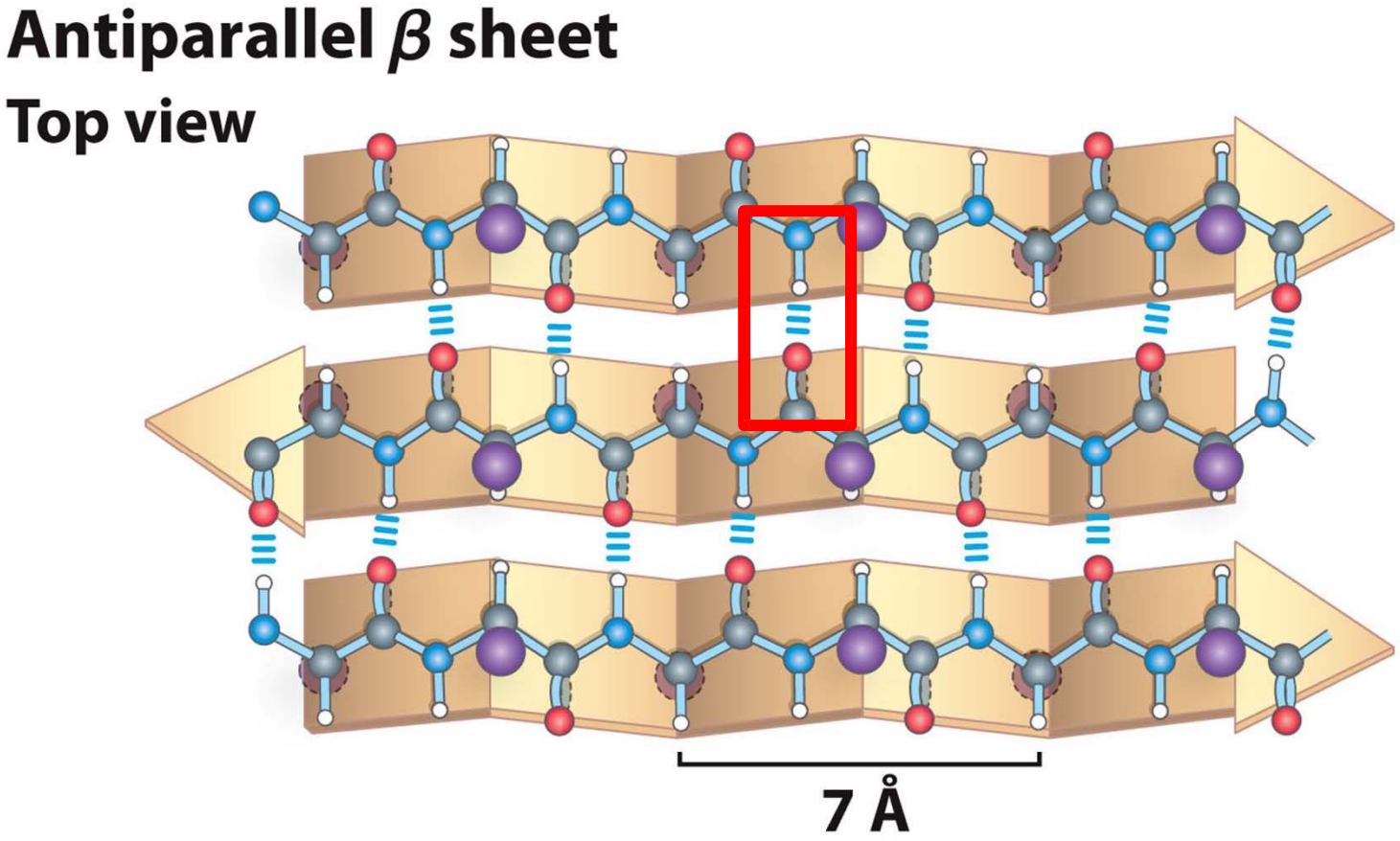


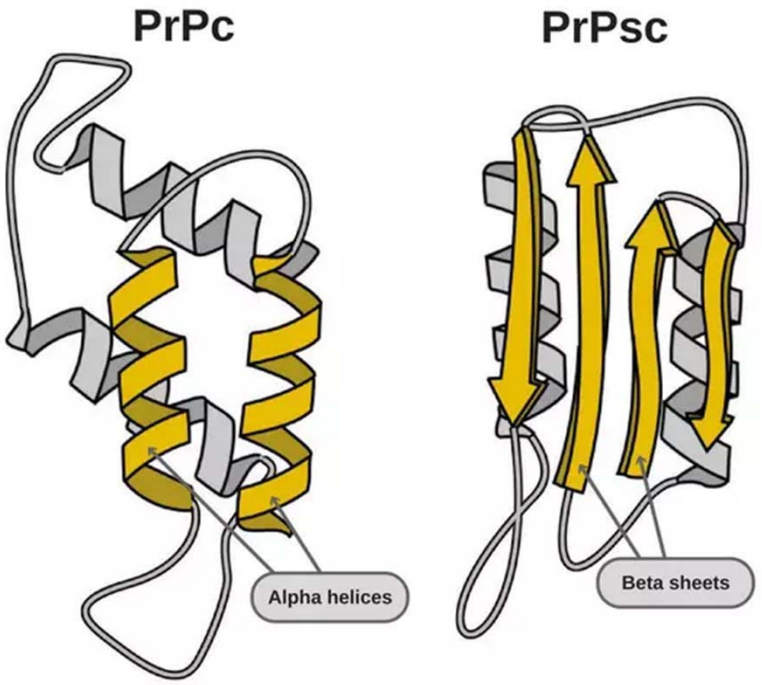
Figure 4-6b
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Prion diseases are caused by misfolding



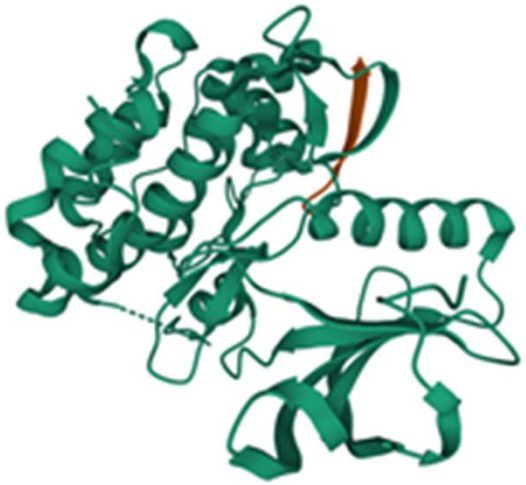
Jmol

PrPSc



The PrPSc and PrPC forms of the protein are different in secondary and tertiary protein structures but the primary structure is conserved. The abnormal isoform contains **beta-sheet conformations** as opposed to the **alpha-helical** folding of the normal protein. The result is a protein which resists degradation by most chemical and physical agents used for sterilization, quite unlike the normal protein. It tends to clump with other PrPSc proteins and strongly bind to the normal prion protein. This induces the misfolding of normal PrPc as well, in a rapidly spreading process which leaves the brain dotted with areas of neuronal loss.

Protein Tertiary Structure

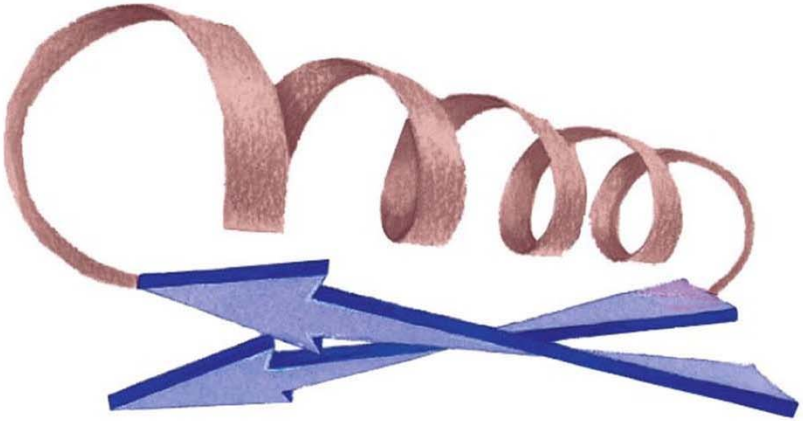


- Tertiary structure refers to the overall spatial arrangement of atoms in a protein.
- Stabilized by numerous weak interactions between amino acid side chains
  - largely hydrophobic, polar and ionic interactions
  - can be stabilized by disulfide bonds
- Interacting amino acids are not necessarily next to each other in the primary sequence.

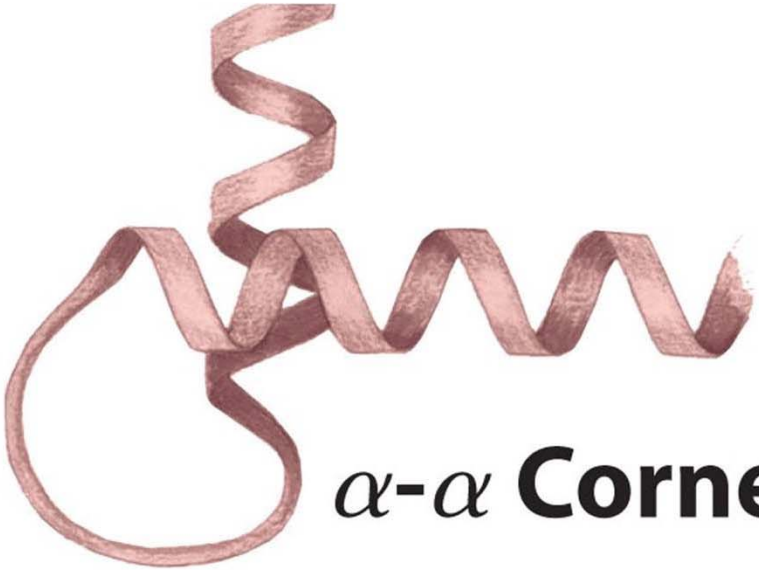
Motifs (folds)

- Specific arrangement of several secondary structure elements
  - all α helix
  - all β sheet
  - both
- Motifs can be found as recurring structures in numerous proteins.
- Globular proteins are composed of different motifs folded together.

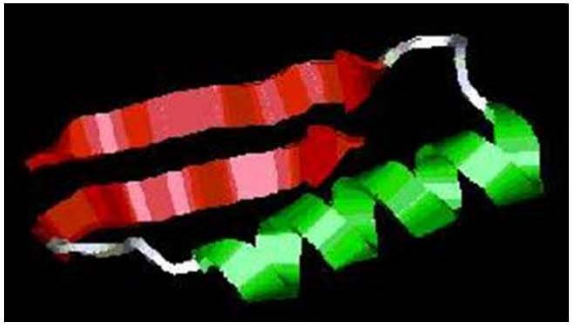
Supersecondary structures

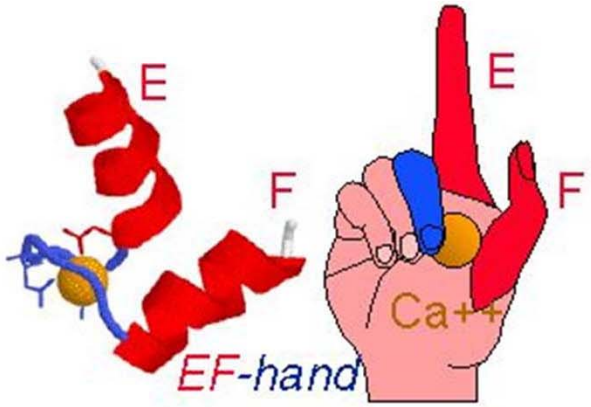


β-α-β Loop



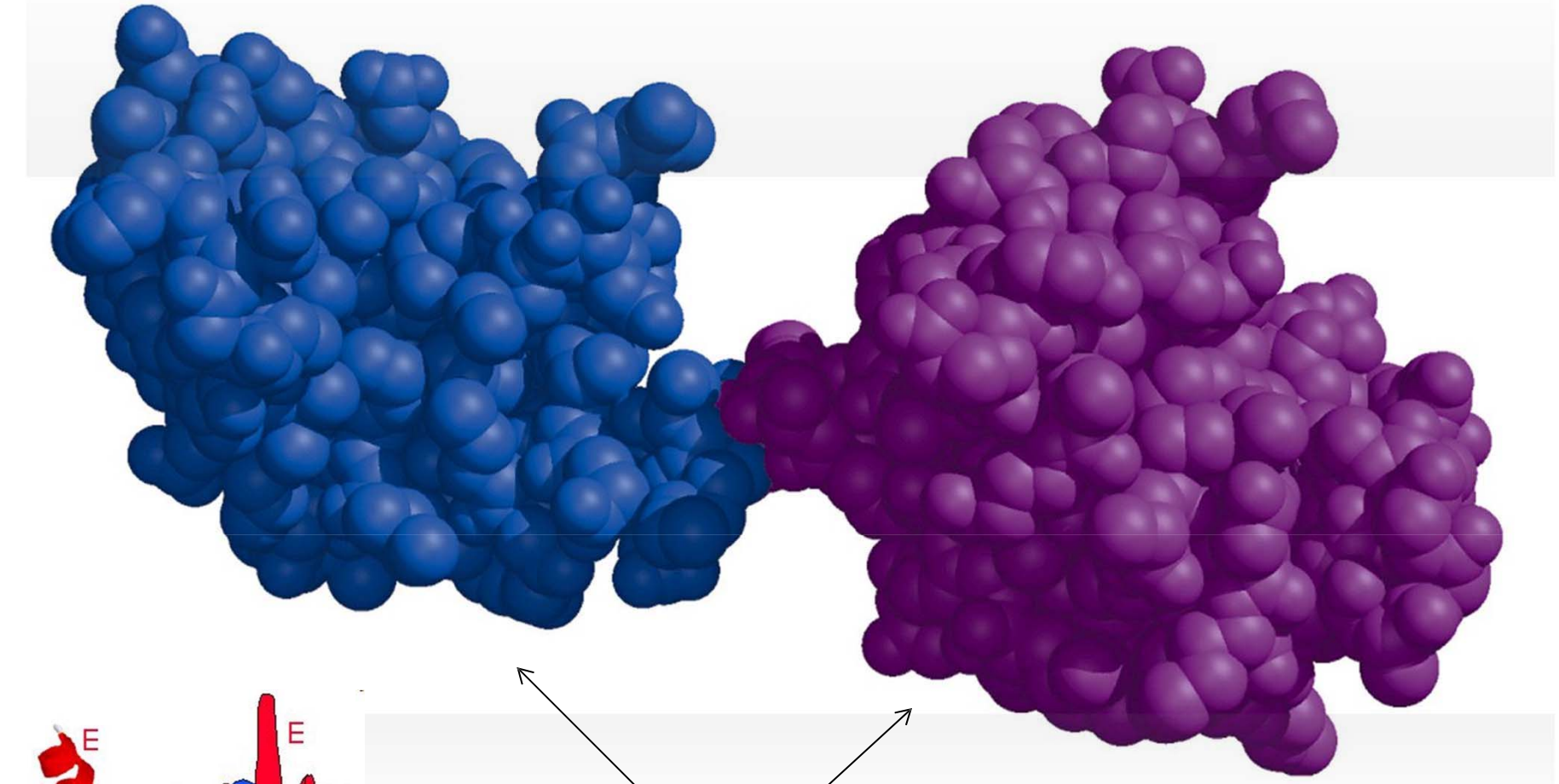
α-α Corner





Domain: is a compact three-dimensional structure with an specific function

Troponin C: is a muscular calcium-binding protein



calcium-binding domains

Structure of Myoglobin

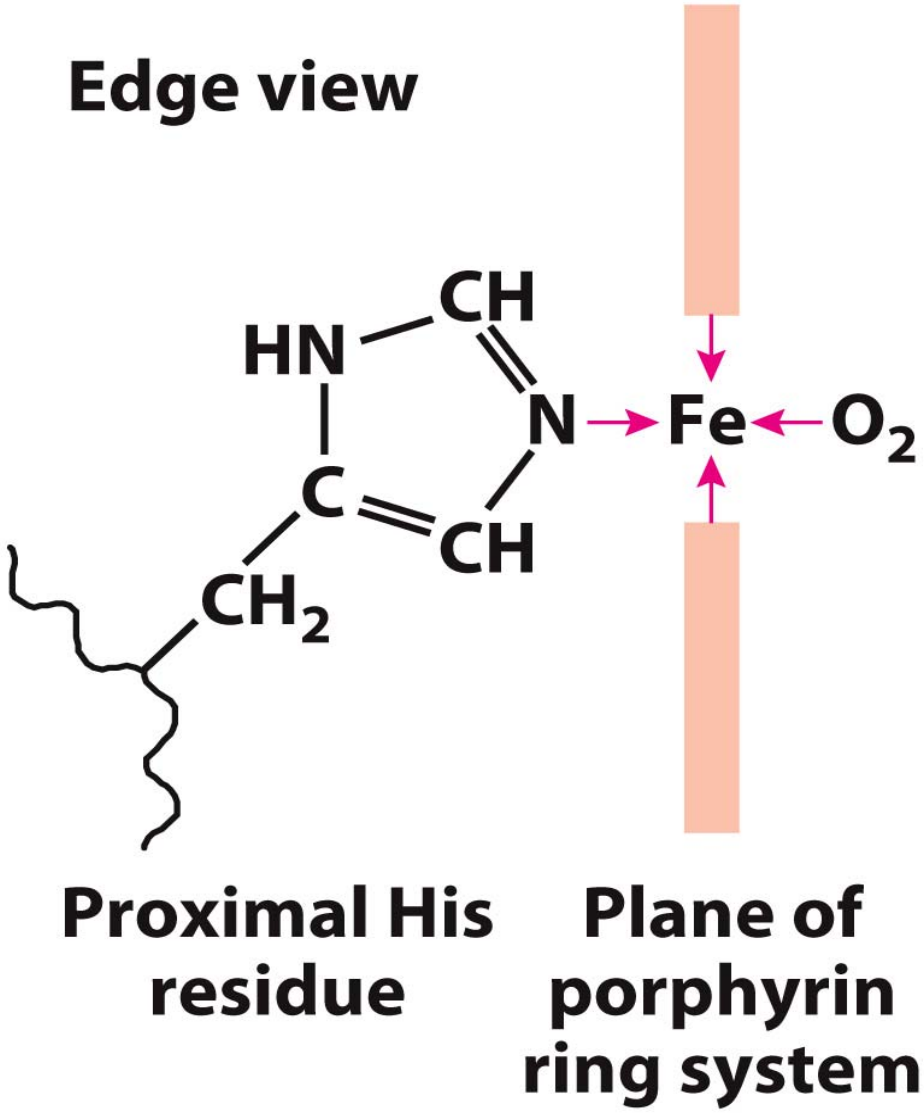


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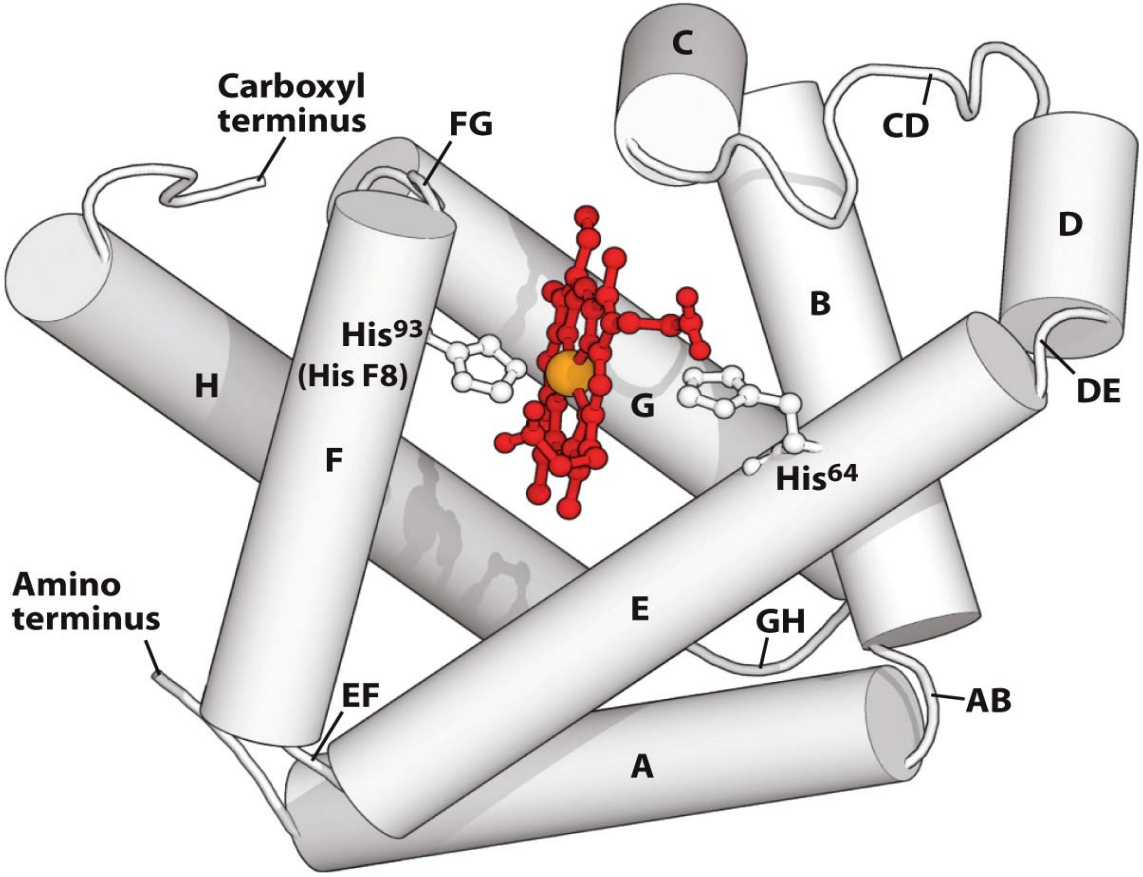


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Quaternary Structure

A quaternary structure is formed by the assembly of individual polypeptides into a larger functional cluster.

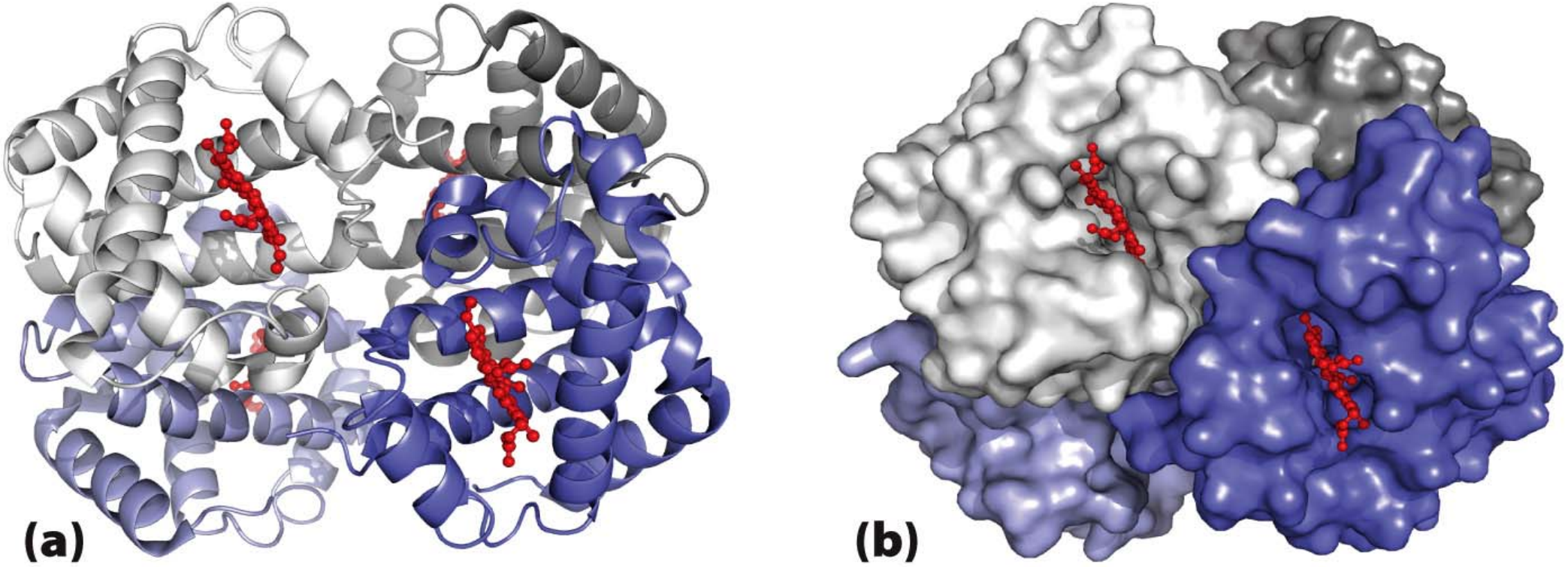


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R and T States of Hemoglobin

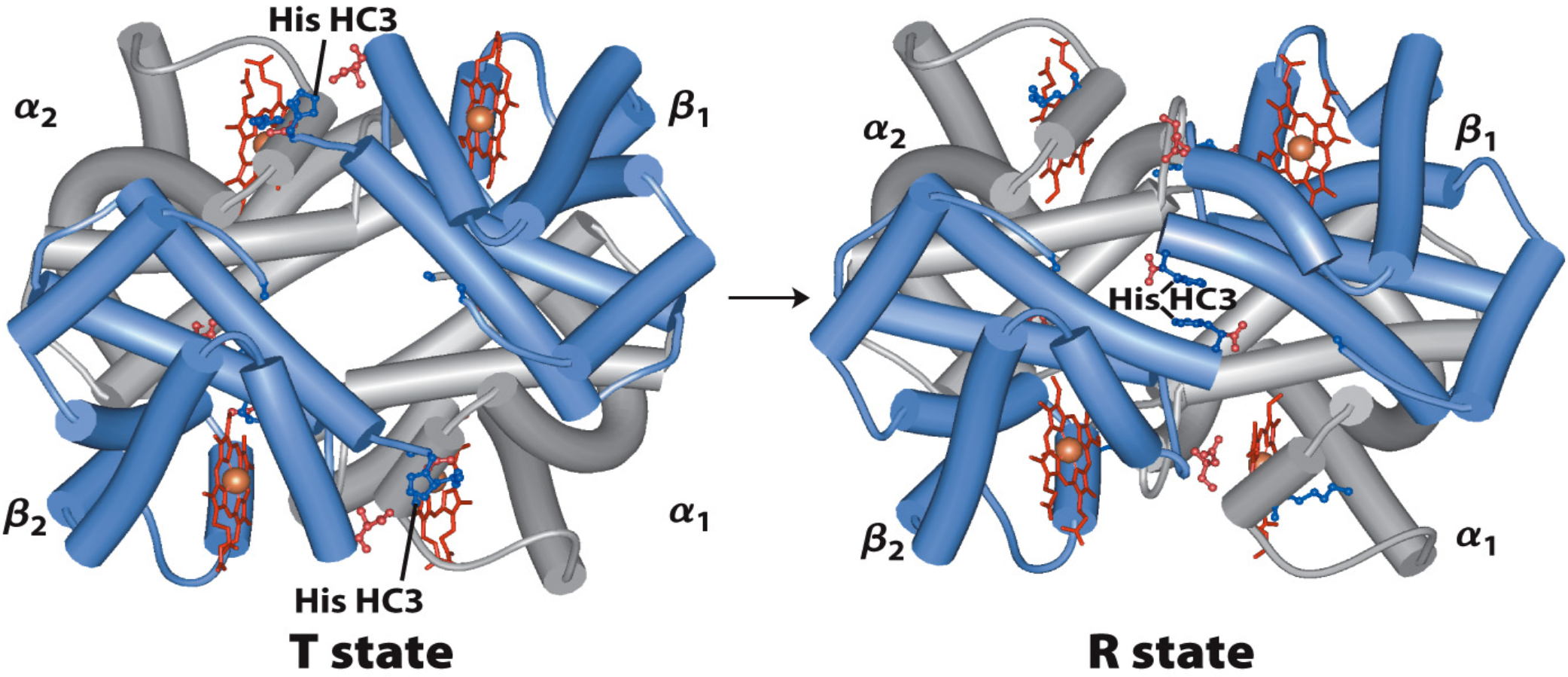


Figure 5-10
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Protein Stability and Folding

- A protein's function depends on its 3D structure.
- Loss of structural integrity with accompanying loss of activity is called denaturation.
- Proteins can be denatured by:
  - heat or cold
  - pH extremes
  - organic solvents
  - chaotropic agents: urea and guanidinium hydrochloride

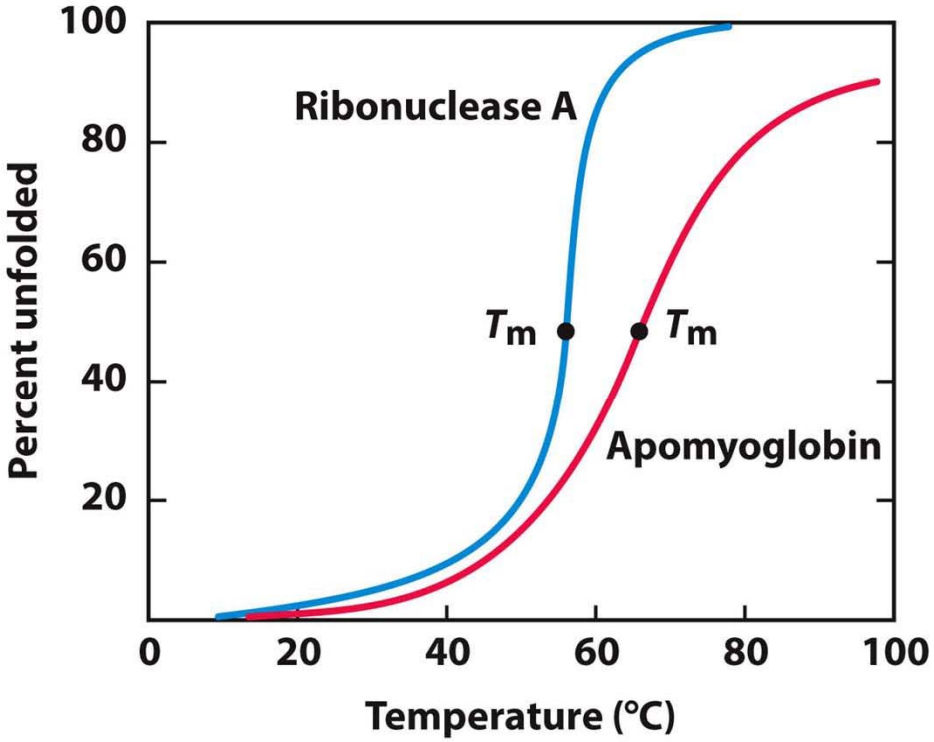


Figure 4-26a
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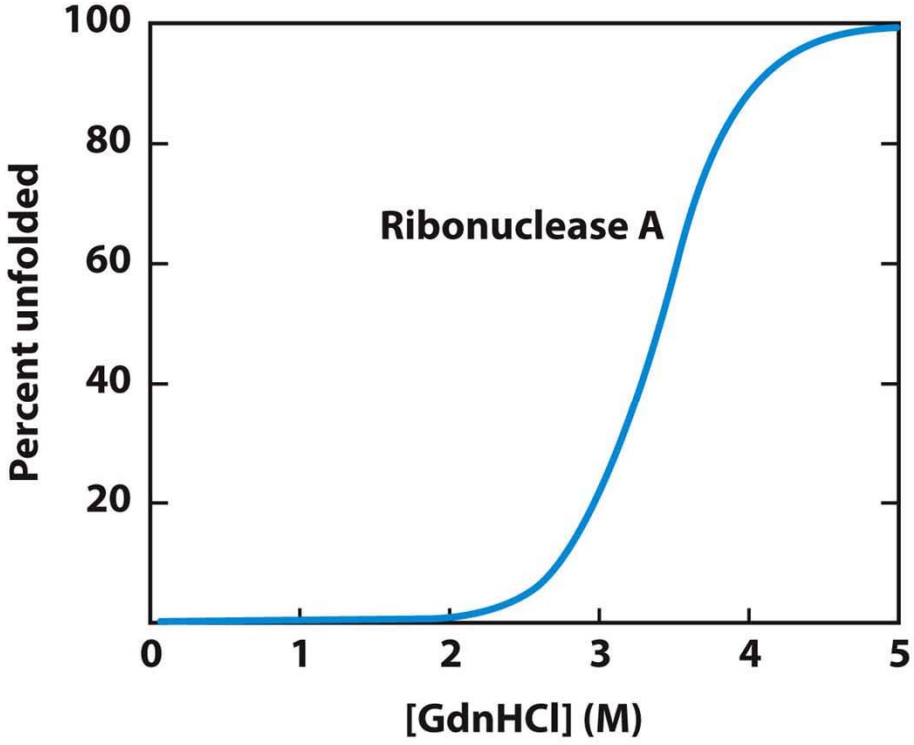


Figure 4-26b
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Have you ever thought how you can make easily a protein denaturation follow up?

Ribonuclease Refolding Experiment

- Ribonuclease is a small protein that contains eight cysteines linked via four disulfide bonds.
- Urea in the presence of 2-mercaptoethanol fully denatures ribonuclease.
- When urea and 2-mercaptoethanol are removed, the protein spontaneously refolds, and the correct disulfide bonds are reformed.
- The sequence alone determines the native conformation.
- The experiment is quite “simple” but so important it earned Chris Anfinsen the 1972 Chemistry Nobel Prize.

Once denaturing agents disappear, **protein refolds** and gets activity.
The questions are: how they do it **so fast** and where is the **information** to obtain always the same conformation?

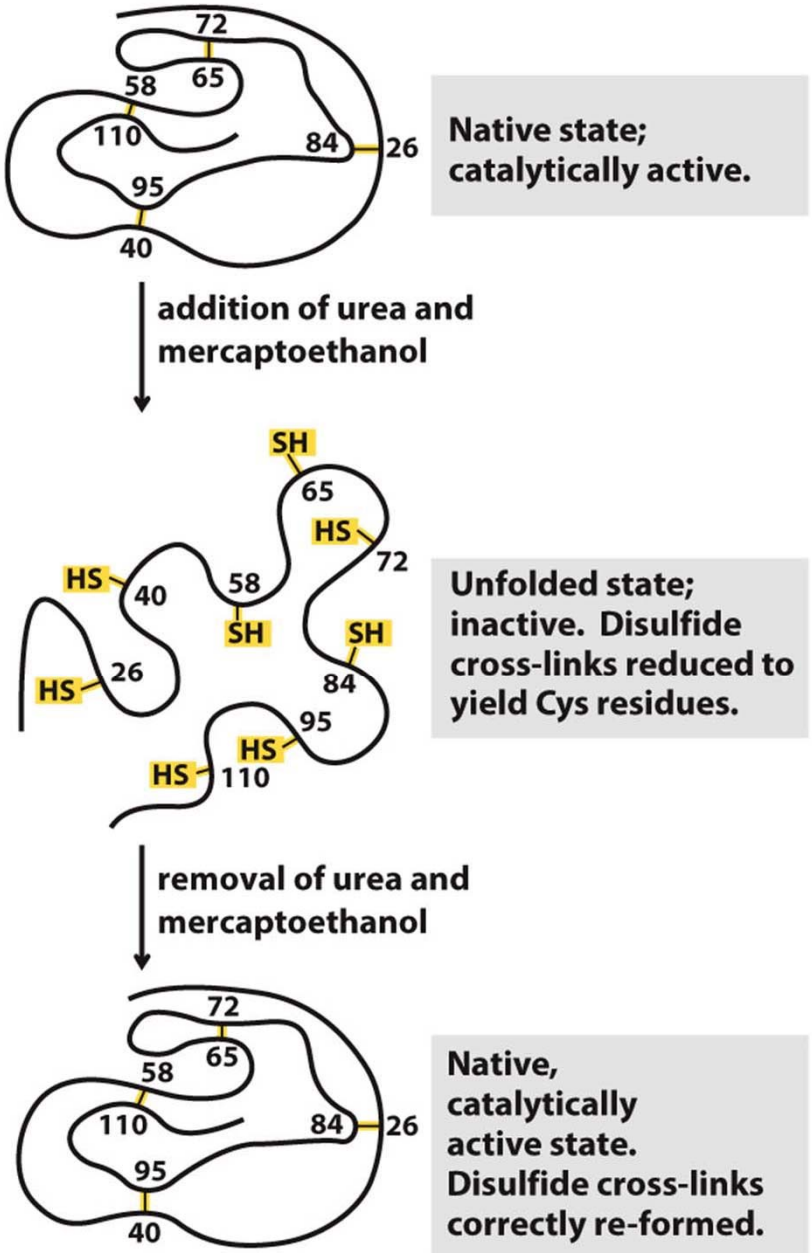


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